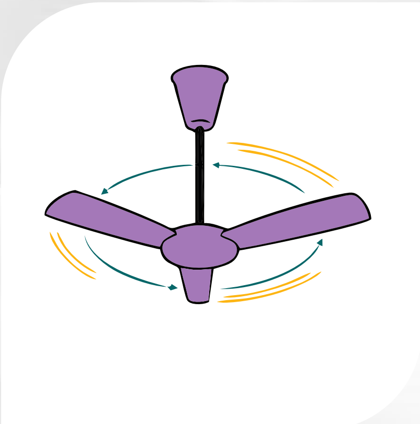
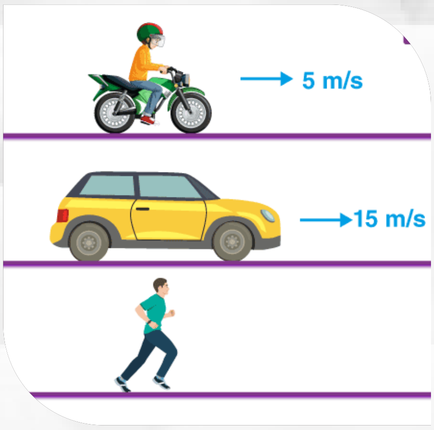
UNIT 02

KINEMATICS





MECHANICS

The branch of physics which is related with the study of motion of objects is called Mechanics. It is divided in two parts: **(i)** Kinematics **(ii)** Dynamics

KINEMATICS

"The branch of mechanics that deals with the motion of objects without Considering the forces that cause motion."

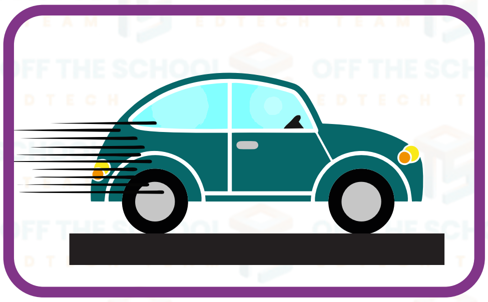
Example

An example of kinematics can be a ball thrown in the air. In kinematics, we would consider the ball's position, velocity, and acceleration at different times during its flight, but we would not consider the forces that cause the ball to move, such as the force applied to it when it was thrown or the force of gravity acting on it.

DYNAMICS

"Dynamics refers to the branch of mechanics that deals with the motion of Objects and the forces that cause the motion."

Example

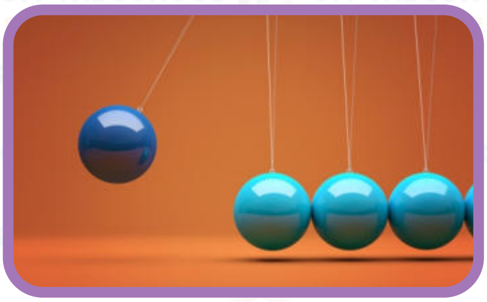


An example of dynamics is a car moving along a road. The car's motion is influenced by various forces such as the engine power, gravity, air resistance, and friction between the wheels and the road

MECHANICS

"Mechanics is the branch of physics that deals with the kinematics and dynamics of objects."

Example



An example of mechanics in action is the motion of a roller coaster. Kinematics is used to describe the position, velocity, and acceleration of the coaster cars as they move along.

REST

"A body is said to be in a state of rest if it does not change its position with respect to its surroundings."

Example

A car parked on the side of the road.
A Book on the table.

MOTION

"A body is said to be in a state of rest if it changes its position with respect to its surroundings."

Example

Child riding a bike.
Women walking on the road.

REST AND MOTION ARE RELATIVE STATES

It is not possible for any object in the universe to be in a state of absolute rest or absolute motion. An object that appears to be at rest relative to one point of reference may also be in motion relative to another point of reference.

Example

1. A person sitting on a train that is moving at a constant velocity, from the perspective of the person on the train is at rest, but from the perspective of an observer standing on the platform, the person is in motion.
2. A car moving at a constant velocity on a highway, from the perspective of a person inside the car, the car is at rest and the landscape is moving. But from the perspective of a person standing on the side of the road, the car is moving.

MOTION

"The motion of an object can be described by its position and change in position over time, speed, velocity, and acceleration."

Types of Motion

There are three basic types of motion:

1. Translatory Motion.
2. Rotatory Motion.
3. Vibratory Motion.

TRANSLATORY MOTION

"A motion in which all points of the moving body move along the same direction is called Translatory Motion."

Types of Motion

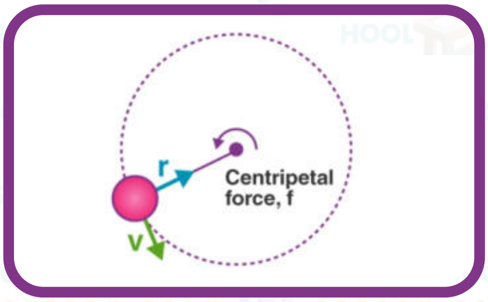
There are three basic types of motion:

1. Translatory Motion.
2. Rotatory Motion.
3. Vibratory Motion.

TYPES OF TRANSLATORY MOTION

a) Linear Motion: The motion of a body along a straight line is called linear motion.

b) Circular Motion:
The motion of a body along a circular path is called circular motion.

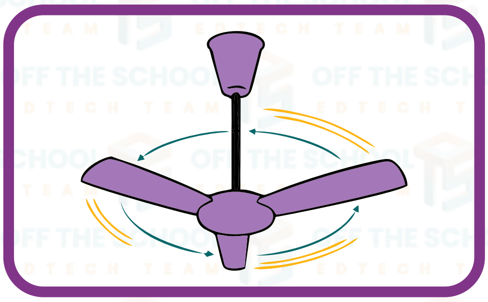


c) Random Motion:
The irregular motion of an object is called random motion.

ROTATORY MOTION

"The motion of a body around a fixed axis that passes through the body itself is known as Rotatory Motion."

Example:
A spinning fan.
Planet orbiting around the sun.



VIBRATORY MOTION

"The repeated movement of an object between two or more positions is called Vibratory Motion. It is also known as Oscillatory Motion."

Example:
Bouncing ball.
Guitar string.

Motion Type	Definition	Example
Translatory Motion	**Movement of an object in a straight line**	A car driving on a highway
Rotatory Motion	**Movement of an object in a circular path around a fixed point**	A Ferris wheel rotating around its central axis
Vibratory Motion	**Back-and-forth movement of an object around a fixed point**	A guitar string vibrating when plucked

DISTANCE

"The total length covered by moving body without mentioning direction of motion".
It is a scalar quantity.
It is denoted by S.
Its S.I. unit is meter (m)
The distance travelled by the person from A to B is either 16 km (purple path) or 24 km (red path).

DISPLACEMENT

"The distance measured in a straight line along a specific direction is called Displacement."
It is a vector quantity.
It is denoted by $\vec{S}$.
Its S.I unit is metre (m).
The displacement of the person is 6 km from A to B due west of A

For example

A person can use three possible paths to move from place A to B. It can be used to illustrate the meaning of distance and displacement. So, if the person takes the purple path, they cover a distance of 16 km from A to B. If they take the red path, they cover a distance of 24 km. However, if the person then moves back from B to A along any of the three paths, the total distance covered would be the same (either 16 km or 24 km), but the net displacement would be zero because the person would be back at their starting point, A

SPEED

Distance covered by an object in a unit time is called speed.

- It is denoted by v.
- It is a scalar quantity.
- S.I unit of speed is metre/second (m/s).

There are many ways to determine speed of an object. These methods depend on measurement of two quantities.
The distance traveled
The time taken to travel that distance
The average speed of an object can be calculated as:

$$\text{Speed} = \frac{\textit{distance traveled}}{\textit{time taken}}$$

$$V = \frac{S}{t}$$

Where **"V"** is the speed of the object, **"S"** distance traveled by it and **"t"** time taken by it.

UNIFORM SPEED

An object covers an equal distance in equal interval of time its speed is known as uniformspeed.

VELOCITY

The rate of change of displacement with respect to time is called velocity.
It is denoted by $\vec{v}$
It is a vector quantity.
S.I unit of speed is meter/second (m/s).
Mathematically;

$$\vec{V} = \overrightarrow{\Delta s}/t$$

Where

$\vec{V}$ = velocity

$\overrightarrow{\Delta s}/t$ = change in displacement

t = time taken

UNIFORM VELOCITY

A body is said to have uniform velocity if it covers equal displacement in equal interval of time in a particular direction.

ACCELERATION

The rate of change of velocity of an object with respect to time is called acceleration.

- It is denoted by $\vec{a}$.
- It is a vector quantity.
- S.I unit of acceleration is meter/Second Square (m/s^2).

Mathematically:

Acceleration = change in velocity/ time taken

$$a = \frac{\Delta t}{\Delta v}$$

$$a = \frac{v\,f - v\,i}{t}$$

When velocity of an object increases or decreases with the passage of time, it causes acceleration. The increase in velocity gives rise to positive acceleration. It means the acceleration is in the direction of velocity. Whereas acceleration due to decrease in velocity is negative and is called deceleration or retardation. The direction of deceleration is opposite to that of change of velocity.

TYPES OF ACCELERATION

There are three types of acceleration.

POSITIVE ACCELERATION

If the velocity of a body increases continuously then the acceleration is said to be positive. It is always in the direction of motion.

NEGATIVE ACCELERATION

If the velocity of a body decreases continuously then the acceleration is said to be negative is called deceleration or retardation. Its direction is opposite to the direction of motion.

UNIFORM ACCELERATION

A constant rate of change of velocity is called uniform acceleration. The uniform acceleration can be calculated by using following formula:

$$\vec{a} = \frac{\vec{vf} - \vec{vi}}{\Delta t}$$

SCALARS AND VECTORS

All physical quantities are divided into two types on the bases of information required to describe them completely.

SCALARS

The physical quantities that have magnitude and a suitable unit are called scalar quantities.
For example:

1. Mass
2. Time
3. Length
4. Distance
5. Work
6. Power
7. Energy **etc.**

VECTORS

The physical quantities which are completely specified by magnitude with suitable unit and particular direction are called vector quantities.
For example:

1. Displacement
2. Velocity
3. Acceleration
4. Force
5. Momentum
6. Torque **etc.**

REPRESENTATION OF VECTOR

Vector diagram is an easy way to represent a vector quantity. The directed line segment can be used to represent a vector. The length of the line segment gives the magnitude of the vector and arrow head gives its direction.

GRAPHICAL ANALYSIS OF MOTION

Graph gives the complete information about the motion of the object based on the measured physical quantities such as distance, speed, time etc.

DISTANCE-TIME GRAPH

A bus travels along a straight road from one bus stop to another bus stop. The distance of the bus from first bus stop is measured every second. The possible motion of the bus as shown by three examples.

GRADIENT

The vertical axis gives rise of the graph while horizontal axis shows its run. The rise divided by run is called **gradient**. When bus travels with uniform speed, the distance time graph is a straight line. When bus travel with non-uniform speed, the distance time graph is a curve. When the bus stops on the next bus stop to drop or pick the passengers the time continues running but the distance stays same. The graph line is parallel to the time axis which shows the bus does not change its position.

EQUATIONS OF MOTION

There are three basic equations of motion for bodies moving with uniform acceleration. These equations are used to calculate the displacement (s), velocity, Time (t), and acceleration (a) of a moving body. Suppose a body is moving with uniform acceleration "a" during some time interval "t" its inertial velocity "vi" changes and denoted as final velocity "vf". The distance "s" covers in this duration of time.

FIRST EQUATION OF MOTION

In the First equation determine the final velocity of a uniformly accelerated body.
Where,

$$\overrightarrow{vi} = initial\ velocity$$
$$\overrightarrow{vf} = final\ velocity$$
$$\vec{a} = acceleration$$
$$\text{t} = \text{time}$$

The average acceleration is the change in velocity over a time interval

$$acceleration = \frac{change\ in\ velocity}{time\ taken}$$

$$\vec{a} = \frac{\overrightarrow{vf} - \overrightarrow{vi}}{t}$$

$$\vec{a}t = \overrightarrow{vf} - \overrightarrow{vi}$$

$$\vec{a}t + \overrightarrow{vi} = \overrightarrow{vf}$$

$$\overrightarrow{vf} = \overrightarrow{vi} + \vec{a}t$$

SECOND EQUATION OF MOTION

The second equation of motion determines the distance covered during some time interval "t", while a body is accelerating from a known initial velocity. As we know, average velocity:

$$\vec{v}\,avg = \frac{\overrightarrow{vf} + \overrightarrow{vi}}{2}$$

$$\vec{v}\,avg = \frac{\overrightarrow{vi} + \overrightarrow{at} + \overrightarrow{vi}}{2}$$

$$\vec{v}\,avg = \frac{\overrightarrow{2vi} + \overrightarrow{at}}{2}$$

$$\vec{v}\,avg = \frac{2\overrightarrow{vi}}{2} + \frac{\overrightarrow{at}}{2}$$

$$\vec{v}\,avg = \overrightarrow{vi} + \frac{1}{2}at$$

Velocity can be given as

$$\vec{v}\,avg = \frac{s}{t}$$

$$\frac{s}{t} = \overrightarrow{vi} + \frac{1}{2}at$$

$$S = (\overrightarrow{vi} + \frac{1}{2}at)t$$

$$S = \overrightarrow{vi}t + {}^{1}\,at^2$$

THIRD EQUATION OF MOTION

Third equation of motion determines the relationship among the velocity and the distance covered by a uniformly accelerated body, where the time interval is not mentioned. Let us take the first equation of motion.

$$\overrightarrow{vf} = \overrightarrow{vi} + \vec{a}t$$

By squaring both sides of the equation we get:

$$(\overrightarrow{vf})^2 = (\overrightarrow{vi} + \vec{a}t)^2$$

We know

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(\overrightarrow{vf})^2 = vi^2 + 2viat + a^2t^2$$

$$(\overrightarrow{vf})^2 = vi^2 + 2a(\overrightarrow{vit} + \frac{1}{2}at^2)$$

By second equation of motion,

$$S = \vec{vi} + \frac{1}{2}at^2$$

$$\overrightarrow{vf}^2 = \vec{vi}^2 + 2aS$$

$$2aS = \overrightarrow{vf}^2 - \vec{vi}^2$$

MOTION UNDER THE GRAVITY

For the motion of bodies under the influence of gravity the equation of motion is slightly modified. Where distance is taken as (s=h) and acceleration is taken as (a=g). Therefore, equation of motion is taken as:

$$\overrightarrow{vf} = \vec{vi} + gt$$

$$S = \vec{vi}t + \frac{1}{2}gt2$$

$$2gh = \overrightarrow{vf}2 - \vec{vi}2$$